BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA

Order Instituting Rulemaking Regarding Transportation Electrification Policy and Infrastructure.	Rulemaking 23-12-008 (Filed December 14, 2023)

COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE ADMINISTRATIVE LAW JUDGE'S RULING INITIATING TRACK 1 AND INVITING PARTY COMMENT

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Date: July 2, 2024

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission ("Commission"), the Vehicle-Grid Integration Council ("VGIC") hereby submits these comments on the *Administrative Law Judge's Ruling Initiating Track 1 and Inviting Party Comment* filed June 3, 2024 by Administrative Law Judges ("ALJ") Marcelo Lins Poirier and Colin Rizzo.

I. INTRODUCTION.

California has set very ambitious transportation decarbonization targets that rely on transportation electrification ("TE") across sectors, with the state requiring all new vehicle sales to be zero-emission by 2036. California's electrification targets for medium- and heavy-duty ("MHD") vehicles are particularly ambitious. On January 1, 2025, Advanced Clean Fleets will begin requiring companies to replace internal combustion vehicles with zero-emission MHD vehicles ("ZEVs"), either based on model year or a milestones option that requires fleets to have a certain percentage of ZEVs. Additionally, even for smaller companies not subject to Advanced Clean Fleets, Advanced Clean Trucks requires ZEVs to make up 5% to 9% of California sales, depending on the truck class. In the light-duty sector, the EV market is more mature and moving beyond early adopters, but access to charging is one of the biggest barriers to EV adoption, especially for renters and people living in multi-unit dwellings ("MUD").

The Commission has repeatedly recognized the critical need to improve access to charging infrastructure; this includes providing ratepayer funding for customers installing EV chargers. In D.22-11-040, the Commission approved an ambitious plan to unify the various charging infrastructure funding programs being executed by the utilities individually and to create a statewide transportation electrification ("TE") Rebate Program. VGIC continues to support this plan and sees benefits in having one program for customers to look to for funding options to support their electrification needs. Without the TE Rebate Program, the Commission risks slowing down EV adoption and, in turn, directly jeopardizes California's TE and broader decarbonization goals. For this reason, VGIC opposes pausing the implementation of the TE Rebate Program. We also offer the following comments below in each of these areas:

- The CPUC's first reason for pausing the program, affordability challenges for ratepayers, contains factual errors:
  - The CPUC fails to consider how vehicle electrification can provide ratepayer benefits in the near term.
  - The CPUC fails to consider how vehicle electrification can provide ratepayer benefits over a longer planning horizon.
- The CPUC's second reason for pausing the program, energization delays due to needed upgrades to distribution infrastructure, contains factual errors:
  - Currently, customers need to purchase chargers for utilities to invest in needed distribution infrastructure.
  - Funding can be leveraged to support sites that do not need or can avoid utility-side infrastructure buildout.
  - Flexible service connection solutions can partially obviate the need for certain distribution system upgrades.

- The CPUC's third reason for pausing the program, the near-term availability of non-ratepayer funds for behind-the-meter transportation electrification infrastructure, contains factual errors:
    - State funding for EV infrastructure is uncertain given the current budget shortfalls.
    - In anticipation of TE Rebate funding availability, critical utility transportation electrification programs were either not fully funded or proposed at all.
- The Commission's approved *VGI strategies adopted per SB 676* depend on the TE Rebate Program.
    - Bidirectional charging equipment should be incentivized through a "V2X adder," for which the TE Rebate Program is the most appropriate venue.
    - Charging sites that promote load during the lowest greenhouse gas emissions hours and lowest utility cost hours should be incentivized through the TE Rebate Program.

II. QUESTION 1: IN LIGHT OF THE CHALLENGES IDENTIFIED ABOVE, SHOULD THE COMMISSION PAUSE IMPLEMENTATION OF THE TE REBATE PROGRAM, INCLUDING ME&O, AND THE LITE PROGRAM? IF NOT, WHY? IF SO, WHY AND WHEN SHOULD PROGRAM IMPLEMENTATION RESUME?

Overall, VGIC does not believe that the TE Rebate Program should be paused due to the following reasons.

a. The CPUC's first reason for pausing the program, affordability challenges for ratepayers, contains factual errors,

i. The Ruling does not consider how integrating electric vehicles into the electric grid can provide short-term ratepayer benefits.

In the Ruling, the Commission highlights "affordability challenges for ratepayers" as a reason for reassessing the TE Rebate program, as Californians are experiencing rapid increases in electric rates.[1] These rate increases are due to critical and necessary investments needed to mitigate

[1] Ruling at 2.

the risk of wildfires, and many of these investments are still yet to be incorporated into rates.[2] In 2022, the required wildfire mitigation investments comprised 8.7 - 17.3% of an average residential bill.[3] Wildfire mitigation investments will continue to be significant in California, and the associated cost recovery is expected to continue driving rate increases for the foreseeable future.

The Commission must consider all potential tools to mitigate ratepayer costs to address the affordability crisis plaguing ratepayers. One critical tool to address ratepayer affordability is vehicle-grid integration ("VGI"). Managed charging and discharging strategies designed to defer or avoid investments in distribution, transmission, and generation resources can help offset the ratepayer cost increases from wildfire-related investments. With this in mind, VGIC strongly urges the Commission to focus on unlocking widespread VGI as soon as possible. As determined by the Commission in D.20-12-029, the near-term benefits of VGI include reducing system peak load and mitigating primary and secondary distribution infrastructure upgrades for new EV charging customers.[4] These ratepayer benefits can be unlocked in the short term by:

1. **Continuing to offer EV customers compelling TOU rate differentials.** Today, California is sending strong time-of-use ("TOU") price signals to encourage charging during off-peak hours - a signal that EVs are following. The use of this "passive" managed charging strategy facilitates a straightforward, accessible customer experience, which is important as many Californians begin driving their first EVs.

[2] Commission, *2023 Senate Bill 695 Report - Report to the Governor and Legislature on Actions to Limit Utility Cost and Rate Increases Pursuant to Public Utilities Code Section 913.1* published May 2023 at p.11-12. Available at: https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/office-of-governmental-affairs-division/reports/2023/2023-sb-695-report---final.pdf
[3] Ibid at p.41.
[4] D.20-12-029 at pages 5, 11, 26-34, at 71-77.

2. **Establishing new managed charging programs targeting distribution system optimization.** While TOU rates encourage system peak load reduction, networked chargers and vehicles can be leveraged to optimize EV charging "behind-the-scenes" in response to local grid conditions. The use of this "active" managed charging strategy leverages mature technologies to facilitate primary and secondary distribution system upgrade deferral, an opportunity that is currently untapped. The (sunsetting) Distribution Investment Deferral Framework ("DIDF") does not address these values and the emergent dynamic rate pilots, including Pacific Gas and Electric Company's ("PG&E") VGI Pilot dynamic rate structure, do not include a dynamic price signal to optimize around these very local grid constraints.
3. **Incentivizing the deployment of bidirectional charging equipment**. For system net peak load reduction or any other managed charging benefits, bidirectional charging equipment offers an opportunity to achieve deeper, more valuable customer response. When coordinating only EV charging load to yield a desired grid benefit, that benefit is limited by the customer's initial plan to charge during a given period. However, leveraging bidirectional charging equipment can provide EV discharge capacity that far exceeds the planned charging capacity. Additionally, the equipment can be operationalized even for EVs that are plugged in but not otherwise scheduled to charge at all. In other words, the maximum EV discharging capability is not limited by a customer's initial plan to charge during a given period, so much larger net peak system load reduction – or distribution deferral benefits – can be achieved. To provide a more specific example, consider an EV charging at 4 kW on a 10-kW charger. A price signal delivered during a system or local peak

could induce a reduction to 0 kW, resulting in a net impact of 4 kW. However, if the 10-kW charger was capable of bidirectional power flow, then during that same period the EV could reduce 4 kW of charging demand *and* discharge 10 kW, resulting in a total impact of -14 kW. This *V2X multiplier* example demonstrates the critical importance of deploying and incentivizing grid-tied bidirectionally-capable charging equipment.

Without TE Rebate Program support for customers, these three strategies to unlock near-term ratepayer benefits will be significantly reduced.

ii. The Ruling does not consider how integrating electric vehicles into the electric grid can provide ratepayer benefits over a longer planning horizon.

Over the long-term, VGIC notes that the increase in electricity sales due to accelerated TE has the potential to lower rates and spread system costs, including critical wildfire mitigation investments, across more kWh sales. The Commission adopted several Findings of Fact detailing that high electricity using customers have contributed more revenue to the grid compared to their costs than low-usage customers.[5] While the creation of an income-graduated fixed charge will reduce this to some extent for residential customers, VGIC notes that encouraging electrified transportation over the next 10-15 years may increase electricity sales and put an overall downward pressure on rates. Early evidence suggests that EVs may already be paying electric bills that in total are higher than their cost to serve in California.[6] Whether this "downward pressure on rates" hypothesis is proven true through existing or forthcoming studies of the previous 5-10 years of investments, the underlying foundation for this concept depends on the extent to which EV

[5] *See e.g.,* D.15-07-001.

[6] Synapse Energy, *Electric Vehicles Are Driving Rates Down for All Customers* published May 2024. Available at: https://www.synapse-energy.com/sites/default/files/Electric%20Vehicles%20Are%20Driving%20Rates%20Down%20for%20All%20Customer%20California%20May%202024%2024-023.pdf

customers charge during off-peak hours and the ability to increase the overall utilization of existing and new grid infrastructure.

With this in mind, rather than asserting whether this downward rate pressure has already materialized from TE efforts in California, VGIC strongly urges the Commission to accelerate efforts that directly support the underlying driver (i.e., shifting EV charging load to off-peak periods). In other words, the Commission can yield a downward pressure on rates by establishing a comprehensive suite of VGI strategies alongside its EV charger deployment efforts. VGIC believes that the share of reliably connected (i.e., "networked") and grid-interactive EVs and EV chargers will increase over the next 10-15 years as customers familiarize themselves not only with the technology but also with the total cost of ownership reductions and revenue-generating opportunities gained from interacting with the grid in new ways (i.e., through VGI programs and rates). However, without statewide TE Rebate support for customers, both near- and long-term ratepayer benefits of VGI efforts will be significantly marginalized as California locks itself into a decelerated TE path.

b. The CPUC's second reason for pausing the program, energization delays due to needed upgrades to distribution infrastructure, contains factual errors.

i. Currently, customers need to purchase chargers for utilities to invest in needed distribution infrastructure.

In the Ruling, the second reason detailing why it may be reasonable to pause the TE Rebate is "energization delays due to needed upgrades to distribution infrastructure."[7] In the Scoping Ruling, the Commission states, "Without adequate utility-side infrastructure, charging ports cannot be energized," and, "A reassessment of the TE Rebate Program would allow the Commission to consider whether ratepayer support for Transportation Electrification should be

[7] Ruling at 2.

focused on utility-side investments in the near-term."[8] VGIC agrees that significant distribution upgrades and utility-side investments will be needed to fully accommodate new TE load, given the ambitious scale of California's TE goals. However, VGIC strongly advises against the Commission pausing TE Rebate funding because of the delay in constructing distribution upgrades.

The Ruling and Scoping Ruling language reveals a concerning "chicken-or-egg" problem: in the Energization Timelines proceeding (R.24-01-018), the utilities have repeatedly stated that they wait to study a project's impact on the distribution system until a new service connection application is completed.[9] Although the utilities state that there are multiple drivers of distribution upgrades, known load requests are consistently cited as significant drivers of individual distribution capacity upgrades.[10] VGIC acknowledges that the Commission is beginning to explore distribution planning process reforms to allow utilities to invest in larger distribution upgrades, like substation upgrades, where load will increase even if it has not materialized yet through new service applications (i.e., "proactive planning and investment").[11] However, these changes will take time to develop, adopt, implement, and refine. While necessary, these process enhancements represent a significant paradigm shift for distribution investment that VGIC anticipates will be subject to challenges from interested parties, further slowing the rollout.

[8] Scoping Ruling at 6.

[9] *San Diego Gas & Electric Company (U 902-E), Pacific Gas and Electric Company (U 39 E), and Southern California Edison Company (U 338-E) Response to Administrative Law Judge Ruling Directing Utility Responses to Questions Regarding Energization Timelines* submitted in Rulemaking ("R.") 24-01-018 on April 22, 2024 at p. 5: "Upstream distribution capacity upgrades, if identified as necessary to accommodate a particular customer's energization request, typically would be initiated in "Step 2: Engineering & Design" of the above-mentioned energization process and should be completed prior to or by the date an energization request is fulfilled." Available at: https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M530/K252/530252757.PDF

[10] PG&E states that currently 28% of substation upgrades and 53% of line/circuit upgrades are caused by an energization request. Ibid at p. 32.

[11] The Commission is currently considering modifying the distribution planning process in R.21-06-017. See the *Administrative Law Judges' Ruling Seeking Comment on Staff Proposal* filed on March 13, 2024 for more detail. Available at: https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M527/K056/527056843.PDF

New service request applications will only be completed and submitted once a customer has concrete plans for the vehicles they are going to electrify and the chargers they will install. Customers installing chargers under Rules 29 and 45 must show proof of commitment to purchase and install the EV supply equipment ("EVSE"), with a purchase order for the chargers being a common way for customers to show their proof of commitment.[12] As a result, deploying chargers and constructing distribution system upgrades are processes that are intertwined and often pursued in parallel, rather than in sequence as suggested by the Ruling and Scoping Ruling.

With this in mind, the funding provided in the TE Rebate program is important to unlock now. Otherwise, customers may delay their plans to electrify their vehicles, which will push out the timelines for utilities identifying known EV load projects and constructing the necessary grid upgrades. Given the inextricably linked nature of this problem, the "chicken-or-egg" must be overcome through *both* the above-mentioned distribution planning process reform and the TE Rebate funding program.

ii. Funding can be leveraged to support sites that do not need or can avoid utility-side infrastructure buildout.

The current slate of utility TE program offers funding support for new service requests related to TE, which may have the effect of nudging customers onto new service drops that can, in turn, trigger upstream distribution system upgrades. However, some customers can achieve their TE goals using existing grid connections and should be encouraged to do so where possible. The TE Rebate Program should be leveraged to support the development of charging sites that do not require utility-side distribution system infrastructure investments. This would further mitigate the

[12] *See* SCE Rule 29 at Footnote 2: "All Applicants, including Applicants that are considered as EVSE manufacturers, a proof of commitment to install the EVSE or Charging Station is required. A proof of commitment is any documentation of clear intent to procure and deploy EVSE, including but not limited to a purchase order, budget approval, grant agreement, request for proposal results, governance body mandated procurement and deployment, approved site plan where the EVSE will be installed, local government permit, etc."

Commission's concerns over ratepayer costs. Additionally, customers may be able to leverage commercially-available technologies reduce peak demand below the full nameplate charging capacity at a site, which could also help avoid utility-side infrastructure buildout. These solutions, which the Commission has referred to as Automated Load Management ("ALM") in D.20-12-029, can support the flexible service connection agreements detailed in Section II.b.iii below. Additionally, if the TE Rebate Program were to support customers that do not trigger a separate service drop for their EVSE, vehicle-to-building ("V2B") use cases would be supported, as detailed below in Section II.d.i.

iii. Flexible service connection solutions can partially obviate the need for certain distribution system upgrades.

Certain customers can energize their charging sites before distribution upgrades are fully constructed by leveraging flexible service connection solutions and agreements. Flexible service connections allow new load to be energized from the grid even if there is not 24/7 grid capacity to accommodate that nameplate capacity for the new EVSE(s) at a site. Relevant bridging strategies include (1) energizing just a portion of the load (i.e., a portion of the chargers), (2) limiting the amount of electricity that can be drawn across the day or year using power control systems or software-based ALM solutions (i.e., allowing for limited or interruptible service), or (3) leveraging temporary DERs to provide behind-the-meter capacity including solar or energy storage when the grid is unable to service the full site. Depending on site characteristics and customer preference, some of these solutions can also be used permanently to avoid distribution upgrades. While the Commission has not established a unified framework for the adoption of flexible service connections, pilots are being conducted by Southern California Edison ("SCE") and PG&E to allow customers to take advantage of these solutions today. Therefore, TE Rebate funding may be used sooner rather than later, obviating the need for the Commission to wait for distribution upgrades before deploying this critical funding source and increasing access to charging equipment.

c. The CPUC's third reason for pausing the program, the near-term availability of non-ratepayer funds for behind-the-meter transportation electrification infrastructure, contains factual errors.

i. State funding for EV infrastructure is uncertain given the current budget shortfalls.

The Commission details a third reason to justify pausing the TE Rebate program: "The near-term availability of non-ratepayer funds for behind-the-meter transportation electrification infrastructure."[13] The Scoping Ruling specifically cites the California Energy Commission's ("CEC") Clean Transportation Program Investment Plan and federal funding that California has received for transportation electrification.[14] However, state-level funding for the CEC's Clean Transportation Program is being reduced due to California's current budget. The CEC had planned to spend roughly $1.5 billion between 2024 and 2028, but $245 million is slated to be cut from this ZEV Package this year.[15] Funding that is planned to be allocated in future years for the CEC is also at risk of being cut since California expects budget deficits to continue into next year and beyond.[16] For this reason, the Commission should not rely on the state budget to provide near-term funding for transportation electrification infrastructure and should continue to leverage ratepayer funding for the TE Rebate Program to ensure California can meet its important and ambitious TE goals.

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[13] Ruling at 2.
[14] Scoping Ruling at 7.
[15] Jessie Gabriel, Chair, Assembly Budget Committee, *Floor Report 2024-2025* published June 22, 2024 at p. 75. Available at: https://abgt.assembly.ca.gov/system/files/2024-06/floor-report-of-the-2024-25-budget-june-22-2024.pdf?utm_source=substack&utm_medium=email
[16] Gabriel Petek, Legislative Analyst's Office, *The 2024-25 Budget: California's Fiscal Outlook* published December 2023 at Figure 4. Available at: https://www.lao.ca.gov/reports/2023/4819/2024-25-Fiscal-Outlook-120723.pdf

ii. In anticipation of TE Rebate funding availability, critical utility transportation electrification programs were either not fully funded or proposed at all.

D.22-11-040 concluded a four-year effort to develop a consistent TE funding approach with the express intention to evolve away from the existing paradigm of ad-hoc, resource-intensive development and litigation of individual make-ready programs. Specifically, the initiative was intended to transition away from "ad-hoc IOU applications that address disparate market segments."[17] For impacted stakeholders, D.22-11-040 marked a strong and unwavering regulatory signal that California was prepared to make the investments required to achieve its TE goals. As a result, utilities, technology providers, site developers, and customers have been awaiting the launch of TE Rebate program.

One direct and palpable impact of the TE Rebate program approval was the withdrawal of PG&E's planned EV Charge 2 program. In 2021, PG&E proposed EV Charge 2 program and detailed a $224 million required budget. In 2022, the Commission authorized PG&E to spend $52 million to implement the EV Charge 2 program.[18] In the end, $52 million proved insufficient and PG&E, with approval from the Commission to choose whether to implement the program, opted to cancel the program before launch.[19] In the decision authorizing EV Charge 2, the Commission stated that it was only approving $52 million in part because PG&E would only need funding as a bridge to the statewide TE Rebate Program and FC1.[20] Given this position, stakeholders have been anticipating the development of the TE Rebate Program.

[17] Transportation Electrification Framework. Energy Division Staff Proposal. February 3, 2020. Page 18. https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M326/K281/326281940.PDF
[18] D.22-12-054.
[19] D.23-09-005.
[20] D.22-12-054 at 19.

Additionally, it is reasonable to expect that PG&E and other utilities could have submitted new TE funding applications but were discouraged due to the forthcoming TE Rebate Program and FC1. As a result, without FC1, no TE funding will be planned for after the existing utility programs (i.e., Funding Cycle 0) expire. In this way, the Commission has created a path dependency; the time window and procedural vehicles to secure funding support for EV customers are now severely limited. If the Commission does pause FC1, guidance on whether the utilities can pursue their own TE and VGI programs via application should be provided.

d. The Commission's approved "VGI strategies adopted per SB 676" depend on the TE Rebate Program

i. Bidirectional charging equipment should be incentivized through a "V2X adder," for which the TE Rebate Program is the most appropriate venue.

Bidirectional charging ("V2X") use cases can leverage the latent battery capacity in EVs to provide backup power to customers, manage customer bills, and support the grid through vehicle-to-grid ("V2G") exports. Currently, V2X systems have a standard interconnection pathway through Rule 21,[21] can receive export compensation through the ELRP,[22] will soon be eligible for shadow bill credits through PG&E's VGI Pilots Phase II (pending approval of PG&E Advice Letter 6909-E) and PG&E and SCE's expanded dynamic rate pilots, and are commercially-available across several customer segments.[23] VGIC commends the Commission for its continued

[21] Specifically, V2G DC EVSE, where the inverter is located inside the charger, are eligible to interconnect under Rule 21.

[22] R.20-11-003. *Phase 2 Decision Directing PG&E, SCE, and SDG&E to Take Actions to Prepare for Potential Extreme Weather in the Summers of 2022 and 2023.* D.21-12-015. https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M428/K821/428821475.PDF. V2X systems can technically receive export compensation through the CEC's Demand Side Grid Support ("DSGS") Program under Option 1 or Option 3. However, this opportunity is limited due to the limited availability of UL 1714 SB-certified V2G DC EVSE. ELRP, PG&E's VGI Pilots Phase II, and SCE and PG&E's Expanded Dynamic Rate Pilots all offer a UL 1741 SB exemption for V2G-DC EVSE. However, although DSGS itself does not require SB certification, the CPUC has not issued guidance related to a UL 1741 SB exemption for DSGS participants, which means DSGS is unlikely to see near-term participation from V2X systems.

[23] *See*, for example: Steve Hanley. Clean Technica. Nissan Using Vehicle to Grid Technology to Power US Operations. November 29, 2018. https://cleantechnica.com/2018/11/29/nissan-using-vehicle-to-grid-technology-to-power-us-operations/; Nuvve Corporation. Blue Bird Delivers North America's First-Ever Commercial Application

support for the V2X market. Real customers have begun to benefit from this support and select instances where the Commission has provided unique provisions – specifically, the dedicated ELRP Subgroup A.5: EV/VGI Aggregations.

Even with these market drivers, high upfront deployment costs remain a critical barrier to the widespread adoption of V2X technology. This is partially due to high interconnection application fees and the incremental cost of bidirectional charging systems, among other factors. To overcome this critical roadblock, VGIC strongly urges the Commission to establish funding to support V2X chargers and associated equipment. Specifically, a V2X rebate adder should be established to partially offset the costs of purchasing and installing V2X equipment that are *incremental* to the costs associated with unidirectional chargers. This V2X rebate adder should take into account the unique needs, technology configurations, and use cases of bidirectional charging technology and not exclude certain technology types or use cases, such as vehicle-to-building ("V2B") installations. Currently, V2B installations are excluded from utility make-ready funding programs because of the requirement that customers install EVSE on a separate service drop to access make-ready funding.

VGIC believes an incremental V2X rebate that covers the additional costs required to deploy a bidirectional charging site is critical to spurring the nascent V2X market in California. Dedicated market transition support and technology deployment incentives have proven effective

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of Vehicle-to-Grid Technology in Electric School Bus Partnership with Nuvve and Illinois School Districts. March 23, 2021. https://nuvve.com/blue-bird-v2g-electric-bus-with-nuvve-and-illinois-school-districts/; Thomas Built Buses / Daimler Trucks North America LLC (2021). The Safe-T-Liner C2 Jouley Electric School Bus. https://thomasbuiltbuses.com/school-buses/saf-t-liner-c2-jouley; Lion Electric. Lion Electric Announces Successful Electric School Bus Vehicle-to-Grid Deployment with Con Edison in New York. December 14, 2020. https://thelionelectric.com/img/medias/LION_Press_Release_White%20Plains%20EN%20FINAL.pdf; Nuvve Corporation (2020). Nuvve DC Heavy Duty Charging Station Specifications Sheet. https://nuvve.com/wp-content/uploads/2020/04/nuvve-dc-heavy-duty-spec-sheet-1.0.pdf; Fermata Energy. Proven Results and Cost Savings with V2G Technology. October 14, 2020. https://www.fermataenergy.com/news-press/proven-results-and-cost-savings-with-v2g-technology; Rhombus Energy Solutions. V2G Charging, Control, and Management 50-500kW: Bidirectional. https://rhombusenergysolutions.com/products; Ford Motor Company. Ford Intelligent Backup Power. https://www.ford.com/trucks/f150/f150-lightning/features/intelligent-backup-power/

in the past. For example, the California Solar Initiative, Self-Generation Incentive Program, utility energy efficiency rebates, and utility TE programs have served as key catalysts for the rapid growth of solar PV, stationary energy storage, energy efficient technologies, and unidirectional charger deployment in California. In turn, these programs accelerated the timeline for realizing the many customer, system, and societal benefits of these technologies. It is reasonable to expect a similar accelerated growth path would be triggered by incentivizing the deployment of V2X charging and associated equipment.

VGIC believes the TE Rebate program is an appropriate venue to offer an incremental incentive to V2X equipment since it is administratively simple, offers reasonable certainty to providers and customers (compared to the limited PG&E VGI Pilots, the only instance VGIC is aware of where V2X charging equipment is offered targeted incentives to support deployment costs), and incorporates consistent treatment across all IOUs in California. VGIC offers itself as a resource and hopes to collaborate closely with stakeholders to determine what incentive level would be appropriate for V2X equipment. At this time, VGIC suggests that a $10 million annual incentive budget carveout in both Funding Cycle 0 (i.e., remaining TE program funding) and FC1 could meaningfully promote BTM V2X equipment in the near-term.

ii. Charging sites that promote load during the lowest greenhouse gas emissions hours and lowest utility cost hours should be incentivized through the TE Rebate Program.

The TE Rebate Program, as adopted in D.22-11-040, would not include support for workplace charging and other non-residential light-duty vehicle ("LDV") serving charging facilities. However, these sites typically experience EV charging during the lowest greenhouse gas emissions hours.[24] These sites can also support charging during the lowest utility cost hours, as

[24] *See*, for example, WattTime signal used in the Self-Generation Incentive Program, which shows GHG emissions are typically lowest during the midday period. https://content.sgipsignal.com/

determined by the Commission's Avoided Cost Calculator. By investing in the development of more daytime charging sites, the Commission can permanently shift load to times with the greatest benefit to ratepayers. Additionally, California is experiencing a large increase of "super commuters" that may require workplace charging before a long journey home (in many cases to AB 841 prioritized communities).[25] With this in mind, VGIC recommends the Commission leverage the TE Rebate Program to support charging sites that can integrate renewable energy generation – a VGI benefit adopted by the Commission in D.20-12-029 – and lower GHG emissions from the grid.

III. WOULD A PAUSE OF THE TE REBATE AND LITE PROGRAMS REQUIRE ANY CLARIFICATIONS REGARDING THE DIRECTIONS, APPROVED BUDGETS, OR OTHER ASPECTS OF THE IMPLEMENTATION OF D.22-11-040? PLEASE EXPLAIN IN DETAIL.

If the TE Rebate program is paused, guidance on how the utilities can seek funding for EV infrastructure should be provided by the Commission as soon as possible.

IV. CONCLUSION.

VGIC appreciates the opportunity to provide this response to the motion. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

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VEHICLE-GRID INTEGRATION COUNCIL

[25] Chris Salviati and Rob Warnock. Apartment List. Explosion of Super Commuters Offers Lessons for Sustainable Growth. (August 16, 2021). https://www.apartmentlist.com/research/explosion-of-super-commuters-offers-lessons-for-sustainable-growth

Date: July 2, 2024